INTERNSHIP REPORT

A report submitted in partial fulfillment of the requirements for the Award of Degree of

Integrated M.Tech in

Computer Science & Engineering
(Computational and Data Science)
by

ANKIT SINGH CHOUHAN

Regd. No.: 22MIP10090

Under Supervision of

Dr. Priscilla Dinkar Moyya

Program Chair

(Duration: 1st Nov to 15th Dec)

SCHOOL OF COMPUTING SCIENCE ENGINEERING AND ARTIFICIAL INTELLIGENCE (SCAI)

VIT BHOPAL UNIVERSITY

December -2025

VIT BHOPAL UNIVERSITY

SCHOOL OF COMPUTING SCIENCE ENGINEERING AND ARTIFICIAL INTELLIGENCE (SCAI)

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CERTIFICATE

This is to certify that the **"Internship report"** submitted by **ANKIT SINGH CHOUHAN (Regd. No.: 22MIP10090** is work done by him/her and submitted during 2025-2026 academic year, in partial fulfillment of the requirements for the award of the degree of **INTEGRATED M.TECH in Computer Science & Engineering (Computational and Data Science)** at VIT BHOPAL.

Internship Guide

PINKY(Director), Sewastack

Program Chair

Dr. Priscilla Dinkar Moyya

SewaStack URN:- UDYAM-UP-18-0071593

To,
Ankit Singh Chouhan
Email: ankitcreativeworks@gmail.com
Phone: +91 78287 36686

December 09th, 2025

Subject: Internship Completion Certificate (Issued Early Upon Request)

This is to certify that Mr. Ankit Singh Chouhan has successfully completed his 1.5 Months internship with SewaStack as a Full Stack Developer Intern from 01st November 2025 to 15th December 2025.

During this internship, Ankit actively contributed to:

- Designing and developing full-stack fetures for web projects
- Creating responsive UI using HTML, CSS, JavaScript, Bootstrap, React, Node.js & Next.js
- Backend development with PHP and MySQL
- Testing, maintaining application functionality, and participating in code reviews
- Following coding best practices and maintaining proper documentation

We appreciate his dedication, professionalism, and enthusiasm throughout the internship. His contributions aligned with the innovative and collaborative culture of SewaStack.
We wish him great success in his future academic and professional endeavors.

Pinky

Warm regards,

Pinky
Director
SewaStack
Email: sewastack@gmail.com
Mobile: +91 89236 98393

ACKNOWLEDGEMENT

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A special note of appreciation is reserved for my supervisor, **Dr. Priscilla Dinkar Moyya,** whose constant support, patience, and expertise were instrumental in completing this project. His guidance, encouragement, and clarity in addressing my queries made this journey both meaningful and enriching.

Lastly, I extend my deepest gratitude to my parents and friends, whose unwavering encouragement and support have been my driving force throughout this research endeavor.
Their belief in me has been a source of strength and motivation.

ANKIT SINGH CHOUHAN

(22MIP10090)

EXECUTIVE SUMMARY / ABSTRACT

1. The Company

SewaStack is a technology-focused software development organization that specializes in building modern, scalable, and user-centric web applications. The company works with contemporary full-stack technologies and emphasizes innovation, clean coding standards, and efficient digital solutions. Its technical environment exposed me to real-world development workflows, professional collaboration, and industry practices, allowing me to gain hands-on experience with both frontend and backend development in a structured environment.

2. The Problem or Opportunity

The internship presented an opportunity to contribute to the design and development of core modules within ongoing web application projects. The organization required improvements in UI responsiveness, backend workflow optimization, and integration of dynamic web components. This opened a space for me to work on real project tasks such as creating responsive interfaces, improving backend logic, resolving performance issues, and enhancing overall user experience. The internship served as a platform to apply theoretical knowledge to practical challenges while contributing to production-level code.

3. Methodology

A structured methodology was followed throughout the internship, beginning with requirement analysis and understanding existing system architecture. Development tasks were executed using agile practices, where work was broken into smaller modules and completed iteratively. My workflow involved designing UI components using React and Next.js, developing backend logic in PHP and MySQL, integrating APIs, testing functionalities, debugging issues, and documenting updates. Tools such as Git, browser developer tools, and MySQL utilities played a crucial role in maintaining high-quality output and ensuring project compatibility across environments.

4. Key Parts of the Report, Findings & Solutions

The report highlights various components of my internship experience, including interface design, API integration, backend module development, performance enhancements, and troubleshooting across multiple layers of the application. My findings revealed that several

system bottlenecks were caused by inefficient queries, inconsistent API responses, and outdated UI structures. The solutions involved optimizing SQL queries, restructuring components for better state management, implementing error-handling mechanisms, and improving front-end responsiveness. These contributions resulted in smoother application performance and a more maintainable codebase for future development.

5. Benefits to the SEWASTACK

My contributions supported faster delivery of UI components, improved reliability of backend modules, and enhanced documentation for ongoing and future development. The optimized database structures reduced query execution time, while improved UI responsiveness led to a better user experience. Additionally, the reusable code components and organized documentation I created helped streamline workflows for other developers. Overall, the work added operational efficiency, reduced debugging overhead, and supported SewaStack's goal of maintaining high-quality, scalable digital solutions.

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LEARNING OBJECTIVES / INTERNSHIP OBJECTIVES

The primary objective of this internship was to provide real-world exposure to software development practices and enable the practical application of concepts learned during the Integrated M.Tech program. The following expanded objectives outline the intended academic, technical, and professional outcomes of the internship:

1. **To gain hands-on experience in full-stack development** by contributing to live projects using technologies such as HTML, CSS, JavaScript, React, Next.js, PHP, and MySQL.
2. **To understand the end-to-end software development lifecycle (SDLC)** including requirements gathering, planning, design, implementation, testing, deployment, and maintenance.
3. **To develop responsive and intuitive user interfaces** following UI/UX best practices, accessibility standards, and component-driven architecture.
4. **To enhance backend development skills** by implementing server-side logic, API integrations, authentication workflows, database queries, and error-handling mechanisms.
5. **To improve debugging and optimization techniques**, identifying and resolving issues related to performance, state management, database latency, and cross-browser compatibility.
6. **To gain exposure to agile methodology**, participating in team stand-ups, sprint planning, module distribution, and iterative development.
7. **To strengthen version control skills** by using Git for branching, merging, reviewing, conflict resolution, and collaborative code management.
8. **To learn professional development practices**, including writing technical documentation, maintaining project structure, and following clean coding standards.
9. **To enhance communication and teamwork skills** by coordinating with developers, understanding feedback, discussing requirements, and participating in code reviews.
10. **To experience real-time project constraints**, such as deadlines, client requirements, design changes, and production-level quality expectations.
11. **To develop analytical thinking and problem-solving skills** through system analysis, workflow understanding, and handling complex development tasks.
12. **To understand database design principles**, including relational schema creation, query optimization, data validation, indexing, and secure data handling.
13. **To observe how industry-grade applications maintain scalability, efficiency, and security**, and to apply these principles to assigned tasks.
14. **To bridge the gap between theoretical concepts and practical implementation**, ensuring readiness for professional software development roles.
15. **To build a strong technical foundation** that contributes to employability, confidence, and preparedness for advanced positions in software engineering.

These objectives collectively guided the structure of the internship and helped maximize the technical, academic, and professional learning experience.

WEEKLY OVERVIEW OF INTERNSHIP ACTIVITIES

1ST WEEK (01 Nov – 07 Nov 2025)

DATE	DAY	DESCRIPTION OF TOPIC / MODULE COMPLETED
01 Nov	Saturday	Attended internship onboarding session, received project introduction, understood company workflow, and set up required tools.
03 Nov	Monday	Studied full project architecture including frontend–backend flow, API structure, and database design. Explored SewaStack development guidelines.
04 Nov	Tuesday	Installed required development software, configured Git & GitHub access, learned branching workflow, and synced local workspace with project repo.
05 Nov	Wednesday	Studied existing UI components, analyzed reusable code logic, and identified improvements in layout structures.
06 Nov	Thursday	Developed initial React components for navigation and UI layout. Focused on modularizing code into smaller reusable parts.
07 No	Friday	Implemented responsive layouts using CSS Flexbox and media queries, tested UI across different screen sizes and browsers.

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2nd WEEK (08 Nov 14 Nov 2025)

DATE	DAY	DESCRIPTION OF TOPIC / MODULE COMPLETED
10 Nov	Monday	Designed homepage layout with dynamic sections. Created wireframes and translated them into functional React components.
11 Nov	Tuesday	Worked on state management using React hooks. Implemented global state for handling navigation and UI changes.
12 Nov	Wednesday	Developed header, footer, navigation bar, and common UI elements ensuring consistency across pages.
13 Nov	Thursday	Integrated frontend with test APIs, handled asynchronous data fetching, processed JSON responses, and rendered data dynamically.
14 Nov	Friday	Debugged visual inconsistencies, aligned UI to company design standards, improved mobile/tablet responsiveness.

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3rd WEEK (15 Nov 21 Nov 2025)

DATE	DAY	DESCRIPTION OF TOPIC / MODULE COMPLETED
17 Nov	Monday	Set up backend development environment for PHP & MySQL. Explored existing backend codebase and server configuration.
18 Nov	Tuesday	Designed relational database schema with tables, foreign keys, and indexes based on project requirement specifications.
19 Nov	Wednesday	Implemented CRUD operations (Create, Read, Update, Delete). Created PHP scripts to interact with the database.
20 Nov	Thursday	Worked on PHP form handling, user input validation, error handling, and sanitization to prevent SQL injection.
21 Nov	Friday	Connected backend APIs with frontend using Axios/fetch. Verified data flow between UI components and backend logic.

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4th WEEK (22 Nov 28 Nov 2025)

DATE	DAY	DESCRIPTION OF TOPIC / MODULE COMPLETED
24 Nov	Monday	Resolved API response delays, added proper error messages, improved backend response formatting.
25 Nov	Tuesday	Implemented user authentication and session management. Worked on login, logout, and credential validation.
26 Nov	Wednesday	Optimized SQL queries using indexing and reduced redundant joins to improve backend performance.
27 Nov	Thursday	Improved dashboard UI with charts, cards, and data visualization components. Refined UI/UX interactions.
28 Nov	Friday	Completed integration of dashboard modules with backend APIs, ensured data refresh and caching worked smoothly.

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5th WEEK (29 Nov 05 Dec 2025)

DATE	DAY	DESCRIPTION OF TOPIC / MODULE COMPLETED
01 Dec	Monday	Conducted UI testing, removed console warnings, optimized React state handling, and cleaned unused code.
02 Dec	Tuesday	Improved performance of heavy components using memoization and conditional rendering.
03 Dec	Wednesday	Added new API routes for handling advanced user operations and updated backend documentation.
04 Dec	Thursday	Prepared module-wise documentation explaining logic, inputs, outputs, and API dependencies.
05 Dec	Friday	Completed debugging cycle, ran functional tests, fixed bug tickets assigned by mentor.

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6TH WEEK (06 Dec 13 Dec 2025)

DATE	DAY	DESCRIPTION OF TOPIC / MODULE COMPLETED
08 Dec	Monday	Enhanced UI/UX interactions and standardized design across screens.
09 Dec	Tuesday	Cleaned backend codebase; removed unused scripts and improved API consistency.
10 Dec	Wednesday	Created final module-wise summary and documentation for submission.
11 Dec	Thursday	Performed final end-to-end testing, resolved compatibility issues.
12 Dec	Friday	Mentor review; applied final corrections and improvements in UI and logic.
13 Dec	Monday	Completed final wrap-up, submitted internship work, and completed exit formalities.

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7TH WEEK (13 Dec 15Dec 2025)

DATE	DAY	DESCRIPTION OF TOPIC / MODULE COMPLETED
15 Dec	Monday	Enhanced UI/UX interactions and standardized design across screens.

Introduction

Internships play a crucial role in bridging the gap between academic learning and professional experience. They provide students with an opportunity to engage with real-world problems, work with industry teams, and put theoretical knowledge into practice. As part of the curriculum of the Integrated M. Tech Computer Science & Engineering (Computational and Data Science) program at VIT Bhopal University, students are required to undergo a structured internship to gain hands-on experience in their respective fields. This internship enables them to apply the skills learned in the classroom to actual industry tasks while gaining exposure to professional workflows, communication standards, project management techniques, and modern technological tools.

The software industry, in particular, demands a strong understanding of both technical concepts and practical implementation skills. The rapid evolution of technologies such as web development frameworks, cloud computing, artificial intelligence, and database systems has created a need for professionals who can adapt quickly and work efficiently with modern tools. To prepare students for such expectations, internships provide exposure to real development environments, agile methodologies, version control systems, live project cycles, and industrygrade coding standards. Through this internship, students not only gain technical proficiency but also understand workplace expectations, teamwork, collaborative development, and clientoriented design processes.

My internship was carried out at **SewaStack**, a dynamic and technology-driven organization specializing in full-stack web development and digital solutions. The company focuses on creating scalable, efficient, and user-friendly applications using modern technologies such as React, Next.js, PHP, MySQL, and JavaScript. With a strong emphasis on clean code, performance optimization, and user experience, SewaStack provided an ideal environment to enhance my development skills and gain industry-level exposure. As a **Full Stack Developer Intern**, I had the opportunity to contribute to ongoing software projects, work closely with experienced developers, and actively participate in both frontend and backend development workflows.

During the course of the internship, I was involved in designing, developing, testing, and optimizing various modules of web applications. The first few days were spent understanding the project architecture, setting up development environments, and exploring the tools and frameworks used by the organization. Once familiar with the workflow, I began working on frontend components using React and Next.js, focusing on building responsive and reusable UI elements. This involved understanding state management, handling dynamic data through APIs, improving layout consistency, and ensuring cross-platform compatibility.

On the backend side, I worked with PHP and MySQL to implement CRUD operations, form validations, database interactions, and secure API endpoints. I also participated in debugging and optimizing backend queries to improve performance and reduce server response times.

This experience gave me deeper insights into how data flows across backend–frontend systems and how different components of a full-stack application communicate with each other. Additionally, I had the opportunity to follow agile development practices, participate in code reviews, write documentation, and understand the importance of clear communication within a development team.

Another important aspect of this internship was the emphasis on problem-solving and independent learning. Throughout the tasks assigned to me, I encountered several challenges such as API integration errors, UI alignment issues, slow SQL queries, and state management bugs. By researching solutions, experimenting with code, and seeking guidance from mentors, I was able to resolve these issues and strengthen my technical confidence. These challenges significantly improved my analytical skills, debugging techniques, and ability to think critically while developing software solutions.

Overall, this internship served as a significant milestone in my academic and professional journey. It enabled me to gain real experience in full-stack development, apply theoretical concepts to real-world projects, and understand the expectations of the software industry. The exposure to professional workflows, advanced tools, and industry standards contributed greatly to my growth as an aspiring software engineer. This report provides a detailed account of the tasks performed, skills learned, challenges encountered, and contributions made during the internship at SewaStack. It also highlights the practical knowledge gained, insights developed, and the overall impact of this internship on my academic and career goals.

INTERNSHIP DISCUSSION

1.Achievement of Internship Objectives

The internship at SewaStack was designed with specific academic and professional objectives in mind, and each of these objectives was systematically achieved through the tasks assigned to me during the internship period. One of the primary goals was to gain real-world exposure to full-stack development. This was successfully accomplished by working on both frontend and backend components of a live project. I gained practical experience working with technologies such as React, Next.js, PHP, MySQL, JavaScript, HTML, CSS, and Git. By participating in actual project development, I learned how to translate theoretical concepts from my Integrated M.Tech curriculum into practical application.

Another key objective was to understand the complete software development lifecycle (SDLC). Throughout the internship, I was involved in various stages such as requirement analysis, UI design, implementation, testing, debugging, documentation, and final review. My involvement in these stages helped me understand how development teams plan projects, divide tasks, communicate progress, and deliver modules within deadlines. The exposure to agile methodologies further strengthened this understanding. Daily interaction with project mentors and regular code reviews helped me appreciate the importance of writing clean, readable, and maintainable code.

The internship also aimed to improve my problem-solving skills. This objective was achieved through several debugging-related assignments, where I was required to identify issues in UI alignment, API communication, backend logic, database queries, and component interaction. Solving these issues helped me develop a structured approach: reproducing the bug, identifying the root cause, applying debugging tools, testing hypotheses, and finally implementing a fix. This strengthened my analytical and critical-thinking abilities.

Another objective was to enhance communication, teamwork, and documentation skills. By participating in team discussions, asking questions, presenting updates, and preparing documentation for completed modules, I improved my professional communication abilities. Overall, all the intended objectives of the internship were fully achieved through consistent involvement in real-world tasks and continuous interaction with mentors and project members.

2. Scientific and Professional Skills Learned During the Internship

The internship offered significant opportunities to learn and refine a wide range of scientific, technical, and professional skills. From a scientific perspective, I gained a deeper understanding of how complex systems interact within a full-stack architecture. I learned the principles of state management in frontend applications, component-based UI development, database indexing, relational schema design, and backend logic structuring. I also developed knowledge of API communication mechanisms, HTTP request handling, data serialization, and integration of frontend and backend systems.

On the technical side, I gained proficiency in React and Next.js, learning how to construct reusable components, handle props and state, implement lifecycle logic using hooks, and optimize rendering performance. I worked extensively with JavaScript, deepening my understanding of asynchronous functions, Promises, and API consumption. On the backend, I learned how to design and implement server-side logic using PHP, work with MySQL databases, write optimized SQL queries, and structure backend modules in a secure and scalable manner. I also learned how to manage sessions, sanitize user input, handle authentication, and structure endpoints for efficient data flow.

I gained professional skills related to version control using Git, such as branching, merging, resolving conflicts, and maintaining commit hygiene. Another important skill was documenting modules, APIs, workflows, and implementation details, which helped in improving clarity for fellow developers. Exposure to agile workflows improved my ability to adapt to iterative development, participate in team discussions, and meet deadlines. Additionally, I developed soft skills such as presentation, written communication, teamwork, time management, and task prioritization.

Working in a professional environment also helped me learn how to interact respectfully with mentors, seek guidance when needed, share progress updates, and respond effectively to feedback. These professional behaviors are essential for long-term success in the software industry.

3. Results, Observations, and Work Experience

The internship produced several tangible results in terms of project contributions, personal improvement, and practical experience. One of the significant outcomes was the development of various frontend modules using React and Next.js. These modules helped improve the user interface of the project, making it more user-friendly, visually consistent, and responsive across different devices. My contributions in designing layouts, handling component interactions, and integrating APIs helped enhance the overall application experience.

On the backend, I implemented functionalities such as CRUD operations, input validation, API routing, and database management. I also worked on optimizing MySQL queries to make the application more efficient. These backend tasks contributed to improving the performance, reliability, and accuracy of data processing within the system.

Through these tasks, I gained the ability to observe how different components of a software system interact with one another. For example, I learned how frontend and backend modules rely on proper API structure, consistent data formats, and synchronized logic to function correctly. This helped me appreciate the importance of coordination between different layers of a full-stack application.

My work experience also included participating in debugging sessions, resolving layout issues, handling state management bugs, and improving API reliability. These tasks required me to analyze code deeply, identify inconsistencies, and apply systematic debugging techniques. Through these experiences, I learned the importance of testing modules thoroughly and writing code that anticipates potential errors or corner cases.

Additionally, I observed how professional teams communicate and collaborate. Discussions during team meetings, code reviews, and technical feedback sessions helped me understand how development decisions are made. I also gained experience in preparing documentation for each completed module, summarizing work done, and maintaining clarity for future developers.

Overall, the internship experience resulted in improved technical depth, enhanced confidence in building real-world applications, and a stronger understanding of full-stack workflows.

4. Challenges Faced During the Internship

Throughout the internship, I faced several challenges that contributed significantly to my learning. One major challenge was understanding the existing project architecture. Working on a live project meant dealing with complex code structures, dependencies, and established workflows. It took time to understand how components communicated, how data flowed through APIs, and how backend logic was structured. Overcoming this challenge required patience, reading documentation, and carefully analyzing different modules.

Another frequent challenge was debugging issues, especially related to frontend behavior, asynchronous requests, and backend data handling. For example, there were moments when API responses did not match expected formats, which caused errors in rendering UI elements. Fixing such issues required analyzing network logs, printing debug statements, testing hypotheses, and verifying backend endpoints. These debugging experiences taught me how to systematically isolate issues and resolve them efficiently.

State management problems in React also presented challenges. Handling dynamic UI updates, ensuring correct data flow, and avoiding unnecessary re-renders required deeper understanding of hooks, component lifecycle, and rendering logic. By experimenting with different approaches and reading official documentation, I gradually became more comfortable with state management techniques.

Backend development also had its challenges. Designing secure and efficient APIs required careful consideration of input validation, error handling, and query optimization. At times, unsafe or inefficient queries caused slow performance or inconsistent results. Learning how to optimize PHP logic and MySQL queries required research, testing, and mentorship support.

Time management was another challenge. Since tasks had to be completed within specific deadlines, I had to plan my work efficiently, break down tasks, and prioritize activities. This helped me develop discipline and accountability in managing project workload.

Despite these challenges, each difficulty contributed to my growth, improved problem-solving skills, and strengthened my technical and professional abilities.

5. Exposure to Industry Tools and Practices

During the internship, I became familiar with various industry-standard tools and practices, including Git version control for code management, Visual Studio Code and browser developer tools for debugging, and project management tools like Jira/Trello for tracking tasks. This exposure helped me understand how professional developers manage code, collaborate on large projects, and maintain project documentation efficiently. It also strengthened my understanding of the importance of clean commits, branch management, and workflow standardization in a team environment.

6. Teamwork and Collaboration

Working in a professional development team emphasized the importance of collaboration. I regularly participated in daily stand-ups, code review sessions, and design discussions. These interactions improved my communication skills, helped me learn how to present technical updates clearly, and taught me the value of giving and receiving constructive feedback. Collaborating with team members also showed me how diverse expertise comes together to solve complex problems and achieve project goals effectively.

7. Contribution to Innovation and Problem Solving

In addition to routine development tasks, I had the opportunity to propose small improvements in UI responsiveness and backend query optimization. Some of my suggestions, such as restructuring frontend components for better modularity and implementing optimized database queries, were adopted by the team. This experience highlighted the value of critical thinking and innovative contributions in a professional setting and encouraged me to proactively suggest solutions when encountering technical challenges.

8. Long-Term Learning and Career Impact

The internship experience significantly influenced my career perspective. I gained clarity on the practical applications of my academic knowledge, learned which areas of full-stack development I enjoy most, and identified skills I need to develop further. This insight will guide my future learning and career choices. Moreover, completing a professional project cycle from start to finish has instilled confidence in my ability to handle complex tasks and adapt to dynamic industry environments, preparing me for future roles in software engineering.

CONCLUSION

The internship at SewaStack has been an invaluable learning experience that significantly contributed to my academic and professional growth. Through hands-on involvement in fullstack development projects, I acquired practical experience in both frontend and backend technologies, including React, Next.js, PHP, MySQL, and JavaScript. The structured tasks enabled me to un

derstand software development lifecycles, project workflows, and the integration of various system components.

During the internship, I achieved the objectives set at the beginning of the program. I strengthened my technical skills, learned industry-standard development practices, improved debugging and problem-solving abilities, and gained exposure to professional collaboration and teamwork. The challenges encountered, such as API integration issues, UI alignment problems, and database query optimization, provided opportunities to enhance analytical thinking and technical competence.

Moreover, the internship helped me develop soft skills such as time management, communication, documentation, and professional etiquette. Participating in team discussions, presenting progress updates, and contributing ideas improved my confidence and ability to work in a collaborative environment. I also learned how to plan tasks effectively and prioritize responsibilities to meet deadlines.

The practical exposure gained from this internship has reinforced my theoretical knowledge, bridged the gap between classroom learning and industry practices, and prepared me for future professional endeavors in software development. Overall, the experience has been enriching, and the insights, skills, and lessons learned will continue to guide my academic and career pursuits.

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